

# Mining Student Behaviour Patterns for Mental Health Insights using Association Rule Learning

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**Abstract**— Mental health problems among university students have become a growing global concern, contributing to poor academic outcomes and increased dropout rates. These issues are shaped by intricate behavioural, financial, and psychosocial interactions that conventional statistical models often overlook. This study applies Association Rule Mining (ARM) using the FP-Growth algorithm in RapidMiner to uncover latent patterns influencing student well-being. Survey data collected from 266 Thai university students were analysed to identify frequently co-occurring behavioural and lifestyle factors. The findings show that family financial support consistently acts as a key protective factor, often linked with adequate sleep and constructive stress-management behaviours. Conversely, digital entertainment—including gaming, social media, and television—emerges as a prevalent coping mechanism associated with both financial stability and sleep quality. Multi-factor rules reveal the interconnected roles of coping style, screen exposure, and body image in shaping mental health outcomes. By offering interpretable, rule-based insights, ARM enables educators and counsellors to design holistic, evidence-based interventions for promoting student mental well-being.

**Keywords**—Mental Health in University Student, Association Rule, Frequent Pattern Growth (FP-Growth), Behaviour and Lifestyle Patterns

## I. INTRODUCTION

By 2025, mental health challenges among university students had escalated into a global concern, primarily driven by mounting academic pressures, financial uncertainty, and the enduring psychological effects of the COVID-19 pandemic. Contemporary studies identify mental health difficulties as a principal cause of student attrition, with nearly 70% of undergraduates reporting psychological distress since the commencement of their academic studies [1]. In the United Kingdom, the prevalence of student mental health problems has increased by approximately 247% since 2015, highlighting the gravity of the situation [2]. Beyond emotional well-being, such conditions markedly affect academic achievement, cognitive functioning, and long-term career prospects [3].

University students in Thailand experience similar mental health challenges. A 2024 survey conducted at Thammasat

University revealed that 72% of students and staff lacked sufficient knowledge of stress management strategies, while more than half exhibited symptoms associated with burnout [4]. Although awareness of mental health issues has improved, access to professional care remains limited and untimely [5].

Existing research on student mental health predominantly depends on survey data and conventional statistical analyses, which often fail to uncover latent behavioural interrelationships. To address this methodological limitation, the present study adopts association rule mining (ARM) to identify significant behavioural correlations associated with mental health. Prior investigations have demonstrated that ARM is effective in detecting clusters of lifestyle-related risk factors—such as smoking, physical inactivity, and insufficient sleep—that are frequently overlooked by traditional analytical methods [6]. In addition, ARM has been utilised within educational research to reveal associations among student behaviours, psychological symptoms, and academic performance [7]. Extending this line of inquiry, the current study examines the influence of sleep quality, family support, screen time, and stress-management practices on student well-being. This data-driven approach demonstrates the potential of ARM to advance mental health analytics by offering interpretable and actionable insights.

Accordingly, this study applies the FP-Growth algorithm, implemented in RapidMiner, in conjunction with ARM to identify frequent behavioural patterns related to student well-being.

## II. RELATED THEORY AND WORKS

This section reviews the theoretical foundations and prior research relevant to the study, including mental health, data mining in educational and health research, frequent pattern growth, and association rule mining.

### A. Mental Health

Universities across the globe are experiencing an increasing prevalence of stress, anxiety, and depression among students, heightening concerns about their mental health. Contributory factors include lifestyle behaviours such as excessive screen time and inadequate sleep, in conjunction

with academic demands and financial hardship [8]. These conditions not only undermine emotional well-being but also adversely affect social interaction and academic achievement [9]. In Thailand, recent investigations indicate similar trends in student stress, burnout, and a lack of awareness regarding effective coping mechanisms [10]. Nonetheless, access to timely mental health support remains insufficient, further hindered by social stigma and limitations in institutional resources [11]. To effectively address these issues, there is a pressing need to investigate innovative and data-driven approaches to mental health analysis and intervention.

### B. Data Mining in Educational and Health Research

Data mining, which encompasses a range of techniques for uncovering latent patterns within complex datasets, has gained increasing significance across both educational and health research domains [12]. In the educational context, methodologies such as association analysis, classification, and clustering have been employed to investigate learning behaviours, identify students at risk, and forecast academic performance [13]. These analytical approaches enable institutions to transcend traditional descriptive statistics by generating actionable insights that facilitate the enhancement of student learning outcomes. Similarly, within health-related research, data mining has been applied to detect behavioural risk factors, categorise mental health conditions, and monitor well-being trends over time [14]. Empirical evidence also highlights its efficacy in revealing interrelationships between mental health and lifestyle determinants, including social support, stress intensity, and sleep quality. Extending these applications to higher education settings presents an opportunity to integrate educational and health perspectives, thereby enriching understanding of how academic engagement and personal behaviours collectively influence students' mental well-being.

### C. Frequent Pattern Growth

Frequent Pattern Growth (FP-Growth) is a widely utilised algorithm for extracting frequent item sets from large-scale datasets. In contrast to the Apriori algorithm, which depends on iterative candidate generation, FP-Growth compresses data into a compact tree structure—known as the FP-tree—and recursively discovers frequent patterns without the need for repeated database scans [15]. Its capability to efficiently process high-dimensional data and uncover latent associations among variables renders it highly suitable for analysing complex domains [16]. Within both educational and health research, FP-Growth has been applied to identify behavioural risk patterns, lifestyle influences, and co-occurring activities. Previous studies have employed this algorithm to examine stress- and well-being-related behaviours, evaluate patterns of student engagement, and determine risk factors associated with suboptimal academic performance. Such applications underscore its capacity to yield meaningful, interpretable results that facilitate evidence-based decision-making [17]. In the domain of mental health, where student data frequently encompass interrelated attributes such as sleep quality, coping mechanisms, and social support, FP-Growth is particularly advantageous due to its efficiency and interpretability. Consequently, the present study adopts the FP-Growth algorithm to identify recurrent behavioural patterns that may

elucidate how students manage stress and sustain mental well-being.

Mathematically, the support of an itemset  $X$  is defined as:

$$\text{Support}(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}} \quad (1)$$

### D. Association Rule Mining

Association Rule Mining (ARM) represents a fundamental data mining technique aimed at discovering meaningful correlations and dependencies within extensive datasets. In contrast to conventional correlation analysis, ARM generates association rules of the form  $\{A\} \rightarrow \{B\}$ , signifying that the occurrence of item  $A$  is likely to be accompanied by item  $B$ . The principal evaluation metrics—support, confidence, and lift—quantify the prevalence, reliability, and relative strength of each rule, respectively [18]. ARM has been widely applied across health and educational research to identify behavioural interrelationships between psychological and academic variables. For instance, a study involving 1,394 Malaysian undergraduates employed the Apriori algorithm to uncover significant associations between stress intensity and mental health status [19]. Likewise, another investigation implemented both positive and negative rule mining to detect strong relationships between depressive symptoms and academic deterioration based on campus-related behavioural and performance data [7]. Building upon these findings, the present research employs ARM to identify patterns in student lifestyle behaviours—such as screen exposure, sleep quality, family support, and stress-management practices—that are linked to mental health outcomes. By generating interpretable and actionable rules, ARM offers a robust analytical framework for guiding targeted interventions and enhancing student well-being initiatives within higher education.

Formally, an association rule is expressed as:

$$X \Rightarrow Y, X, Y \subseteq I, X \cap Y = \emptyset \quad (2)$$

where  $I$  represent the set of all possible items in the dataset,  $X$  and  $Y$  are itemset, which are subsets of  $I$ .

The support of a rule is defined as:

$$\text{Support}(X \Rightarrow Y) = \frac{|\{t \in T: X \cup Y \subseteq t\}|}{|T|} \quad (3)$$

where  $T$  is a collection of individual transactions,  $t$  represents a single transaction within the transaction collection  $T$ .

The confidence of a rule is defined as:

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)} \quad (4)$$

The lift of a rule is given by:

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)} \quad (5)$$

### III. METHODOLOGY

This section outlines the methodology adopted in this study, structured as follows:

#### A. Conceptual Framework

The motivation for this study arises from the demand for transparent and actionable insights into student mental health. Although predictive machine learning models can achieve high accuracy in identifying at-risk students, they frequently function as “black boxes,” providing limited explanation of the behavioural relationships underlying their predictions. This lack of interpretability reduces their practical value for educators and mental health practitioners, who require clear and contextually grounded explanations based on student experiences.

The first step in the process is collecting survey data, which includes behavioural, lifestyle, financial, and psychosocial aspects of student life. Each variable is transformed into a binary indicator to prepare the dataset for mining. This creates a transaction-like structure in which every student record is viewed as a group of “items.” After then, FP-Growth is used to identify frequently occurring item sets, which are behaviours or conditions that occur together after a certain support threshold. From these item sets, association rules are developed and evaluated using metrics on lift, confidence, and support.

This framework as shown in Fig. 1. prioritizes uncovering patterns rather than making predictions. It provides an approach for investigating individual risk factors and the related relationships that influence students' well-being. By emphasizing the understanding of student's pattern of behavioural, lifestyle, financial, and psychosocial aspect, the framework offers practical insights for universities and mental health professionals, facilitating the creation of interventions that tackle interrelated elements of student life instead of addressing risk factors separately.

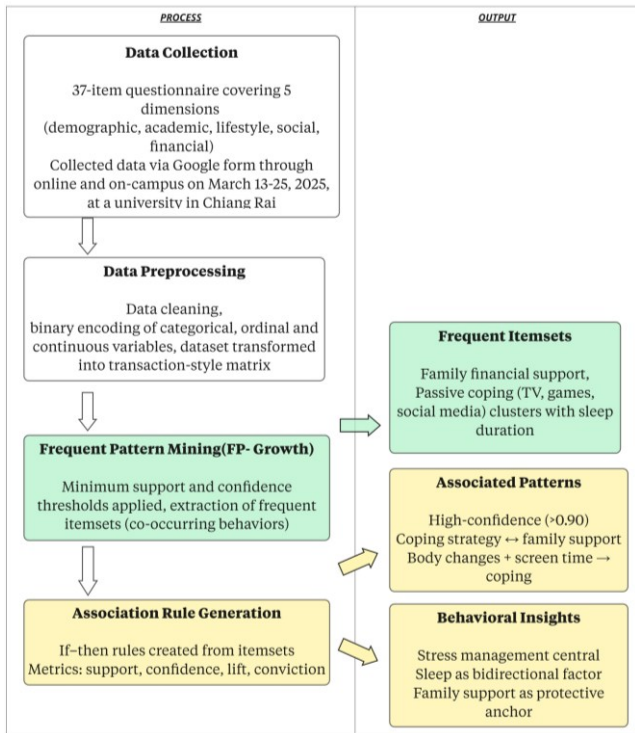


Fig. 1. Research framework

#### B. Data Collection

Collecting actual behavioural and lifestyle data from university students to explore patterns related to mental health was the goal of the data collection process. To reduce stigma, encourage participation, and cover a broader range of student experiences, this study intentionally focused on non-clinical data rather than clinical approaches that use diagnostic instruments like the PHQ-9 or GAD-7. A solid Google Form with 37 questions was developed using the findings of previous research on student well-being [20] [21] [22]. The questions were organized into five thematic areas: demographics, academic achievement, lifestyle habits, social involvement and financial factors.

To ensure a representative and diverse population, a mixed data collection strategy was used. About two-thirds of the answers were gathered in-person in public spaces on campus, while the remaining responses were gathered online via university groups and social media. This combined approach improved the response rate and decreased potential sample bias. Over a period of a week in March 2025, the survey was conducted at a university in Chiang Rai, and 266 valid responses were obtained. Participants came from a variety of academic fields and ranged in age from 18 to 25. The survey was brief, anonymous, and voluntary, with informed consent provided at the beginning.

In addition to academic and lifestyle questions, the survey assessed stress management techniques and discriminatory experiences (e.g., based on physical appearance or gender identity). Furthermore, students were asked to use a 5-point Likert scale to evaluate their overall condition of mental health. Although self-reported information may lead to inflated or deflated results, this method facilitated extensive involvement and scalability. Significantly, the dataset is not designed for clinical diagnosis but to detect behavioural patterns that could be linked to mental health risk

#### C. Data Preparation

The collection of 266 student survey answers was methodically prepared to guarantee its appropriateness for association rule mining. All attributes were transformed into binary indicators that indicate the presence or absence of specific behaviours or situations because the FP-Growth algorithm needs a transaction-style structure. This method enabled each student's record to be viewed as a set of items for analysing frequent patterns.

Data cleaning was the initial phase, which involved removing unnecessary metadata, such as timestamps. Only variables associated with academic workload, lifestyle decisions, coping mechanisms, financial circumstances, and self-reported mental health were preserved, while demographic factors were omitted to maintain concentration on behavioural and psychosocial aspects.

All categorical features were transformed into binary values. Additionally, ordinal items that were evaluated using a five-point Likert scale were transformed into a binary format. While the answers of 4 or 5 were represented as 1, suggesting greater intensity or agreement, responses of 1, 2, or 3 were categorized and recorded as 0, indicating low to moderate levels. This modification made a sharper distinction between less and more noticeable behavioural patterns.

Midpoint thresholds based on the survey ranges were used to discretize continuous variables such as sleep duration,

weekly study hours, and exercise frequency. Values at or above the midpoint were represented by 1; values below the midpoint were represented by 0. This method provided a clear distinction between the student group's above-average and below-average behavioural levels. These modifications made it possible for the FP-Growth algorithm to generate association rules and extract frequently occurring item sets in a way that is both computationally efficient and comprehensible for further analysis.

#### D. Analytical Process in Rapid Miner

The analytical process was created in RapidMiner Studio to find frequent item sets and produce association rules from the processed dataset. After importing the cleaned survey data, the Select Attributes operator was employed to keep only the behavioural, lifestyle, financial, coping, and psychosocial variables. These characteristics were transformed into dummy-coded binary format using the Numerical to Binominal operator, allowing the dataset to be handled in a transaction-style manner.

Common patterns were identified utilizing the FP-Growth operator. The minimum support was established at 0.8, ensuring items sets present in at least 80% of student records were preserved while eliminating uncommon or insignificant combinations. This limit enabled the examination to focus on patterns that were both common and significant within the sample. The item sets were subsequently utilized as input for the Create Association Rules operator, with confidence selected as the primary assessment criterion. A confidence level of at least 0.70 was utilized to ensure that only reliable and stable rules were produced. Extra metrics like lift and conviction were calculated to reflect the strength of relationships beyond mere random co-occurrence.

This process produced two complementary outcomes: frequent item sets indicating groups of co-occurring student attributes, and association rules formulated as understandable if-then statements. Collectively, these results formed the basis for recognizing behavioural patterns associated with student mental health, which are detailed and analysed in Section IV.

### IV. RESULTS

This section demonstrates the results of the study which comprises of four sections: FP Growth, Association Rules, Behavioral Insight and Implications, and Comparison with previous research.

#### A. FP Growth

The FP-Growth identified several highly frequent behavioural and lifestyle patterns among university students (as Table I) with the minimum support is set to 0.8. Finance family support appeared to be the most frequent factor (support = 0.895), highlighting its important role in student well-being. Other significant single-item patterns included managing stress by entertainment (support = 0.695) and average hours of sleep (support = 0.648), indicating that digital stress coping mechanisms and sleep quality were dominating aspects of student life.

The behavioural patterns are likely to occur together, reflecting underlying behavioural and contextual factors among students. For example, the co-occurrence of managing stress through TV, games, or social media and average hours

of sleep highlights a group of students who balance stress management with adequate sleep. Similarly, the three-way itemset of finance family support, managing stress through digital entertainment and average hours of sleep with a support of 0.421 shows that these patterns often cluster together, forming a consistent lifestyle pattern among the students.

TABLE I. FREQUENT ITEM SETS IDENTIFIED BY FP-GRWOTH

| No. | Size | Frequent itemset                                                                                               | Support |
|-----|------|----------------------------------------------------------------------------------------------------------------|---------|
| 1   | 1    | Finance family support                                                                                         | 0.895   |
| 2   | 1    | Managing stress (watch TV/ play games/ or scroll social media)                                                 | 0.695   |
| 3   | 1    | Average hours of sleep per night                                                                               | 0.647   |
| 4   | 2    | Finance family support, managing stress (watch TV/ play games/ or scroll social media)                         | 0.643   |
| 5   | 2    | Finance family support, Average hours of sleep                                                                 | 0.579   |
| 6   | 2    | Managing stress (watch TV/ play games/ or scroll social media), Average hours of sleep                         | 0.462   |
| 7   | 2    | Average hours of sleep, Screen time impact                                                                     | 0.361   |
| 8   | 3    | Finance family support, managing stress (watch TV/ play games/ or scroll social media), Average hours of sleep | 0.421   |

#### B. Association Rules

Parameter setting for extracting the associated rule are defined as 1) criterion = "confidence", minimum confidence = "0.7", gain theta = "2.0", and Laplace K = "1".

Table II represents the association rules discovered between pairs of factors, showing the relationships between them and their corresponding support, confidence, and lift values. These rules provide insight into how specific behavioural patterns are interconnected and linked within the student population.

TABLE II. THE TWO-ITEM ASSOCIATION RULES

| No. | Rules ( $A \rightarrow B$ )                                                                                                          | Support | Conf | Lift |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|---------|------|------|
| 1   | Managing stress (watch TV/ play games/ or scroll social media) $\rightarrow$ finance family support                                  | 0.64    | 0.92 | 1.03 |
| 2   | Managing stress (talk to friends or family) $\rightarrow$ Finance family support                                                     | 0.43    | 0.95 | 1.06 |
| 3   | Impact of Screen Time on Concentration and Productivity $\rightarrow$ Managing stress (Watch TV/ play games/ or scroll social media) | 0.39    | 0.72 | 1.04 |
| 4   | Unexplained body changes since starting university $\rightarrow$ Managing stress (watch TV/ play games/ or scroll social media)      | 0.39    | 0.71 | 1.03 |
| 5   | Satisfaction with Physical Appearance $\rightarrow$ Finance family support                                                           | 0.37    | 0.90 | 1.01 |
| 6   | Used medications pain relievers $\rightarrow$ Managing stress (watch TV/ play games/ or scroll social media)                         | 0.32    | 0.77 | 1.11 |
| 7   | Confidence in managing academic workload $\rightarrow$ Average hours of sleep per night                                              | 0.31    | 0.74 | 1.14 |

For example, the first rule (Managing stress by TV/games/social media → Finance family support) has support of 0.64, indicating that it applies to 64% of students. Its confidence of 0.92 shows that almost every student who manages stress by digital entertainment also reports family financial support, and the lift of 1.39 confirms the positive and strong association between the two.

Similarly, the second rule (Managing stress talk to friends or family → Finance family support) shows a slightly lower support of 0.43 but the higher confidence of 0.95 means that although fewer students use this stress-coping mechanisms overall, when they do, there is an even stronger likelihood that they also receive family support.

Rules three and four demonstrate links between personal factors and stress management behaviours. Specifically, students who experience screen time impacts on concentration (support = 0.39, confidence = 0.72) or physical body changes since starting university (support = 0.39, confidence = 0.71) tend to manage stress through digital leisure activities. These connections highlight the influence of lifestyle and physical conditions on stress-coping choices.

In addition, rule five (Satisfaction with physical appearance → Finance family support) shows a moderate relationship (support = 0.37, confidence = 0.90), suggesting that students who feel positive about their appearance often also perceive financial stability within their families. Rule six (Used medications for pain relief → Managing stress by TV/games/social media) presents support of 0.32 and confidence of 0.77, identifying a subgroup of students who pair medical coping with leisure-based methods. Finally, rule seven (Confidence in managing academic workload → Average hours of sleep per night) shows a support of 0.31 and a relatively high lift of 1.14, indicating that students who effectively manage their academic demands are more likely to maintain healthier sleep patterns.

TABLE III. THE THREE-ITEM ASSOCIATION RULES

| No. | Rules (A, B → C)                                                                                                                                      | Support | Conf  | Lift |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|------|
| 1   | Managing stress (watch TV/ play games, or scroll social media), Average hours of sleep per night → Finance family support                             | 0.42    | 0.91  | 1.01 |
| 2   | Finance Family Support, Impact of Screen time on Concentration and Productivity → Managing stress (watch TV/ play games/ or scroll social media)      | 0.36    | 0.756 | 1.08 |
| 3   | Managing stress (watch TV/ play games/ or scroll social media), Managing stress (talk to friends or family) → Finance Family Support                  | 0.29    | 0.97  | 1.09 |
| 4   | Average hours of sleep per night, Unexplained body changes since starting university → Managing stress (watch TV/ play games/ or scroll social media) | 0.27    | 0.74  | 1.07 |
| 5   | Finance family support, Satisfaction with physical appearance → Average hours of sleep per night                                                      | 0.27    | 0.73  | 1.14 |

Table III presents the rules involving three factors. The first rule shows that students who cope with stress through

leisure activities such as TV, gaming, or social media, combined with reporting their average hours of sleep, are also highly likely to receive family financial support (support = 0.42, confidence = 0.91, lift = 1.01). Likewise, the second rule reveals that students with financial support and high screen-time impact tend to engage in leisure-based stress management (support = 0.36, confidence = 0.756, lift = 1.08). The third rule emphasizes that combining multiple coping strategies—both digital leisure and talking with friends or family—is strongly associated with reporting family support (support = 0.29, confidence = 0.97, lift = 1.09). The fourth rule shows that students experiencing body changes since freshman year, along with sleep habits, often also rely on stress management through leisure (support = 0.27, confidence = 0.74, lift = 1.07). Finally, the fifth rule indicates that students with both family financial support and satisfaction with their physical appearance tend to report adequate sleep (support = 0.27, confidence = 0.73, lift = 1.14).

### C. Behavioral Insight and Implications

The association rules reveal that students' behaviours, coping strategies, and well-being factors are deeply interconnected rather than isolated. Stress management surfaces as a central theme, whether through digital platforms, it consistently links to other aspects of student life such as financial support, sleep patterns, or physical changes. This suggests that the way students handle stress is often influenced by structural support and health-related conditions.

Sleep also plays a crucial role, it emerges both as an antecedent and as a consequence in several rules, spotlighting its bidirectional relationship with stress, confidence and self-image. Adequate sleep seems to be both an outcome resulting from factors such as family support, satisfaction with appearance and a resource that shapes coping behaviours.

Financial support shows up as a returning anchor in high-confidence rules. This indicates that financial stability from family serves as a foundation upon which other coping mechanisms and well-being behaviours are developed. In contrast, students without this support may face challenges, as financial security limits their ability to sustain leisure and healthier routines.

### D. Comparison with previous research

This research builds on earlier work predicting student mental health with machine learning models like Random Forest, XGBoost, and deep learning, which identified key factors such as academic performance, sleep deprivation, family support, and passive coping. It highlighted that excessive screen time, low workload confidence, and weak academic support are strongly linked to mental health issues.

This study uses Association Rule Mining (ARM) to reveal behavioural patterns not shown by predictive models. Unlike prior work, ARM offers clear, rule-based insights into how factors like stress, sleep, and family support interact in real life, such as "Stress via TV/Games + Sleep → Family Support."

Additionally, ARM examines behaviours like screen time, coping, and body image, showing how they often occur alongside protective factors like sleep and family support. This complements previous clustering studies that classified at-risk students as either socially adaptive or isolated profiles.

Together, these studies provide two perspectives: one showing behaviour patterns and their links, and the other highlighting key factors affecting mental health. This comprehensive approach offers a deeper understanding of student well-being and guides targeted support in educational settings.

## V. CONCLUSION

This study demonstrated how the FP-Growth algorithm, combined with association rule mining, can provide useful insights into the behavioural patterns associated with the mental health of students. Instead of focusing prediction accuracy as traditional machine learning techniques do, the analysis highlighted co-occurring factors like sleep, coping strategies, and family support, presenting them via transparent if-then rules. These rules demonstrate how common actions overlap, providing a solid foundation that universities can convert into proactive support measures.

However, there are limitations to the study. External validity was limited by the dataset's participation of only 266 students from a single university. Recall and social desirability bias might have an impact on self-reported metrics. Additionally, the cross-sectional approach captures simultaneous occurrences at a single moment but fails to clarify how these behaviours and conditions evolve or influence each other over time. Lastly, the results' high frequency of financial support emphasizes the frequency of this component appears in the dataset, potentially overshadowing less common but noteworthy patterns.

In order to improve generalizability, future research needs to fill these gaps by expanding data collecting across multiple universities and using bigger sample sizes. Longitudinal studies and the use of data from digital wearable devices, such as screen-time records or sleep trackers might provide more objective and evolving insights. Comparative studies using different association rule methods and hybrid strategies that merge interpretable rules with predictive modelling could improve both robustness and applicability.

In conclusion, this study provides a comprehensive, pattern-based framework for understanding the well-being of students. Transitioning from black-box predictions to interpretability grounded in rules establishes a basis for developing early-warning tools that are both data-driven and accessible to educators, counsellors, and policymakers.

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